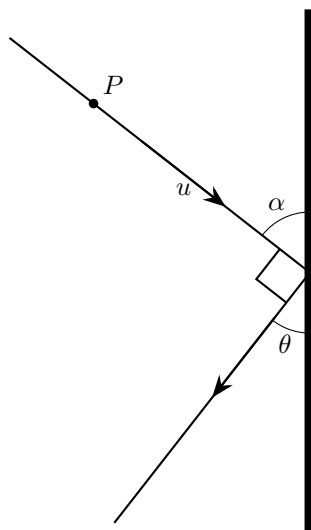


**Instructions**

Unless a question states otherwise, take  $g = 9.8 \text{ m s}^{-2}$  wherever a numerical value of  $g$  is required.

**Questions**

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1. A particle  $P$  of mass  $m$  is moving with speed  $u$  on a fixed smooth horizontal surface. It collides with a fixed smooth vertical barrier. Before impact its path makes an angle  $\alpha$  with the barrier, and after impact its path makes an angle  $\theta$  with the barrier. The incident path and the outgoing path are perpendicular. The coefficient of restitution between  $P$  and the barrier is  $e$ . The particle loses 36% of its kinetic energy in the collision.

Find the value of  $e$  and the exact value of  $\tan \alpha$ .

[5]

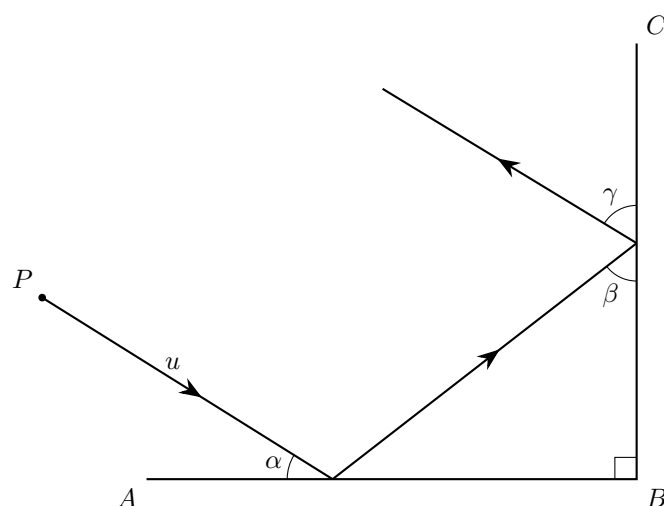
- 2.** A hockey puck moving on a smooth horizontal rink collides with a fixed straight side-board. Let  $\mathbf{i}$  be a unit vector parallel to the side-board and let  $\mathbf{j}$  be a unit vector perpendicular to the side-board, directed away from it.

Immediately before the collision, the puck has velocity  $(9\mathbf{i} - 12\mathbf{j}) \text{ m s}^{-1}$ .

Immediately after the collision, the puck moves with speed  $10 \text{ m s}^{-1}$  in the direction of the vector  $3\mathbf{i} + 4\mathbf{j}$ .

Model the puck as a particle.

- (a) Find the coefficient of restitution between the puck and the side-board. [3]
- (b) Determine whether or not the side-board is smooth. [3]  
Fully justify your answer.



3. The smooth walls  $AB$  and  $BC$  are at right angles to each other. A particle  $P$  moves on a smooth horizontal floor with speed  $u$  and strikes the wall  $AB$  at an angle  $\alpha$  to  $AB$ . It rebounds, then strikes the wall  $BC$  at an angle  $\beta$  to  $BC$ . After the second impact it rebounds at an angle  $\gamma$  to  $BC$ , as shown in the diagram. The coefficient of restitution between  $P$  and each wall is  $e$ .

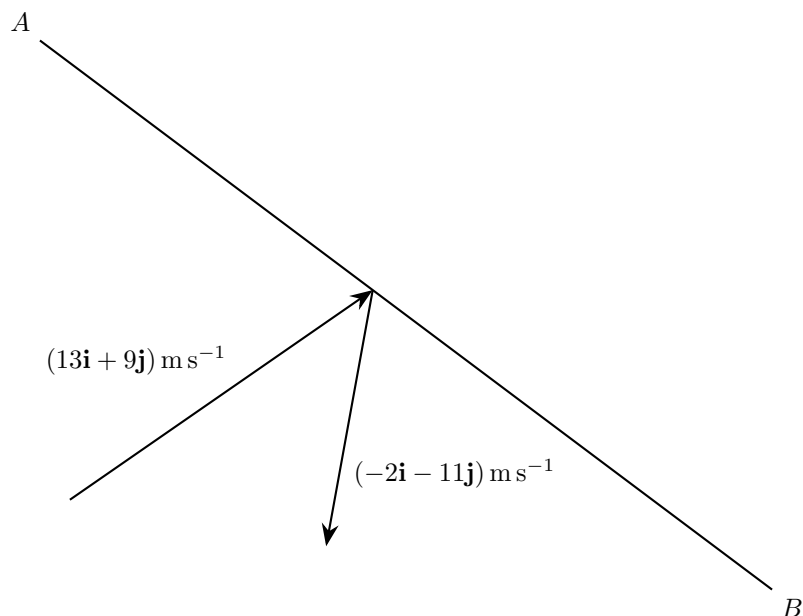
(a) Show that

$$\tan \alpha \tan \beta = \frac{1}{e} \quad [3]$$

(b) Express  $\gamma$  in terms of  $\alpha$  and explain what this implies about the final direction of motion of  $P$ . [4]

During the first impact the particle loses four times as much kinetic energy as it loses during the second impact. After the second impact the kinetic energy of  $P$  is  $\frac{9}{25}$  of its initial kinetic energy.

(c) Find the value of  $e$  and the value of  $\tan \alpha$ . [4]



4. The diagram above shows the plan view of part of a smooth horizontal table. The line segment  $AB$  represents a fixed smooth vertical wall.

A small ball of mass  $0.4 \text{ kg}$  moves across the table and rebounds after striking the wall.

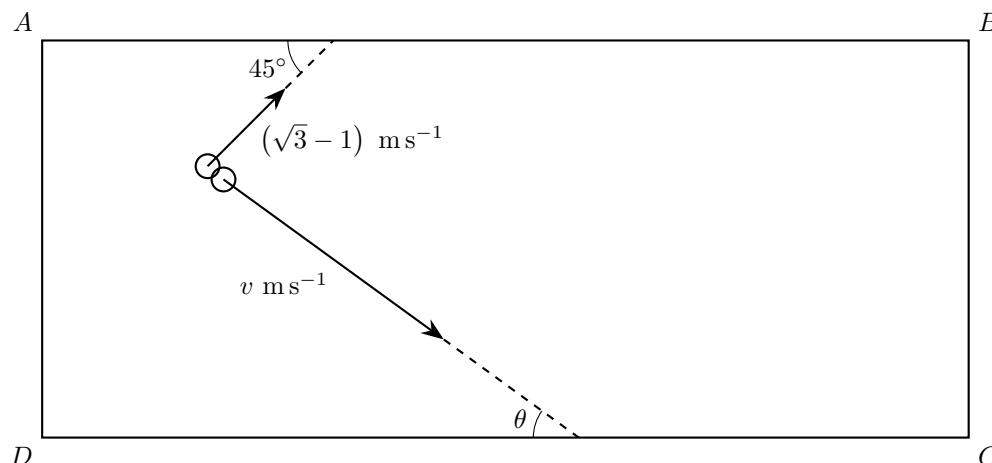
Immediately before impact its velocity is  $(13\mathbf{i} + 9\mathbf{j}) \text{ m s}^{-1}$ .

Immediately after impact its velocity is  $(-2\mathbf{i} - 11\mathbf{j}) \text{ m s}^{-1}$ .

The coefficient of restitution between the ball and the wall is  $e$ .

(a) Show that  $AB$  is parallel to  $(4\mathbf{i} - 3\mathbf{j})$ . [4]

(b) Find the value of  $e$ . [5]



5. Two snooker balls, one white and one red, have equal mass.  
The balls are on a horizontal table  $ABCD$ .

The white ball is struck so that it moves at a speed of  $2 \text{ m s}^{-1}$  in a direction making an angle of  $15^\circ$  with  $AB$ , towards  $CD$ .

The white ball hits a stationary red ball.

After the collision, the white ball moves at a speed of  $(\sqrt{3} - 1) \text{ m s}^{-1}$  and at an angle of  $45^\circ$  to  $AB$ .

After the collision, the red ball moves at a speed  $v \text{ m s}^{-1}$  and at an angle  $\theta$  to  $CD$ .

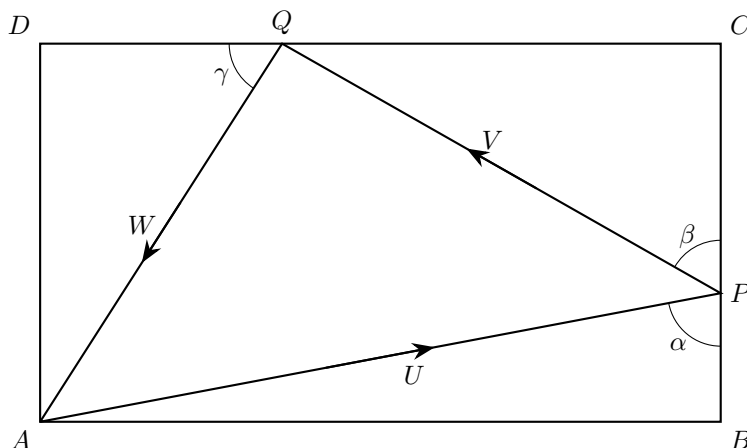
When the collision takes place, the point of impact is twice as far from  $CD$  as it is from  $AB$ .

The diagram above shows the velocities of the balls after the collision.

After the collision, the white ball hits  $AB$  and the red ball hits  $CD$ .

Model the balls as particles and neglect any resistance.

- (a) Explain why the two balls hit the sides of the table at the same time. [2]
- (b) Show that  $\theta = 36.2^\circ$  correct to one decimal place. [4]
- (c) Find  $v$ . [2]
- (d) Determine which ball travels the greater distance after the collision and before hitting the side of the table.  
Fully justify your answer. [2]
- (e) State one possible refinement to the model that you have used. [1]



6. A small smooth billiard ball is projected from the corner  $A$  of a horizontal rectangular table  $ABCD$ .

The ball first hits the side  $BC$  at the point  $P$ , then hits the side  $CD$  at the point  $Q$ , and then returns to  $A$ .

Angle  $APB = \alpha$ , angle  $CPQ = \beta$  and angle  $DQA = \gamma$ .

The ball moves along  $AP$  with speed  $U$ , along  $PQ$  with speed  $V$  and along  $QA$  with speed  $W$ , as shown in the diagram.

The coefficient of restitution between the ball and side  $BC$  is  $e_1$ .

The coefficient of restitution between the ball and side  $CD$  is  $e_2$ .

The ball is modelled as a particle.

(a) Show that  $\tan \beta = e_1 \tan \alpha$ . [4]

(b) Hence show that  $e_1 \tan \alpha \tan \gamma = e_2$ . [3]

(c) By considering  $\angle PAQ$  or otherwise, show that it is possible for the ball to return to  $A$  only if  $e_2 > e_1$ . [6]

If instead  $e_1 = e_2$ , the ball would not return to  $A$ .

(d) Given that  $e_1 = e_2$ , use the result from part (b) to describe the path of the ball after it hits  $CD$  at  $Q$ , explaining your answer. [1]

7. A particle  $P$  is moving with velocity  $(7\mathbf{i} - 4\mathbf{j}) \text{ m s}^{-1}$  on a smooth horizontal plane. It collides with a smooth vertical wall which is parallel to the direction vector  $2\mathbf{i} + \mathbf{j}$  and rebounds with velocity  $(2\mathbf{i} + 6\mathbf{j}) \text{ m s}^{-1}$ .

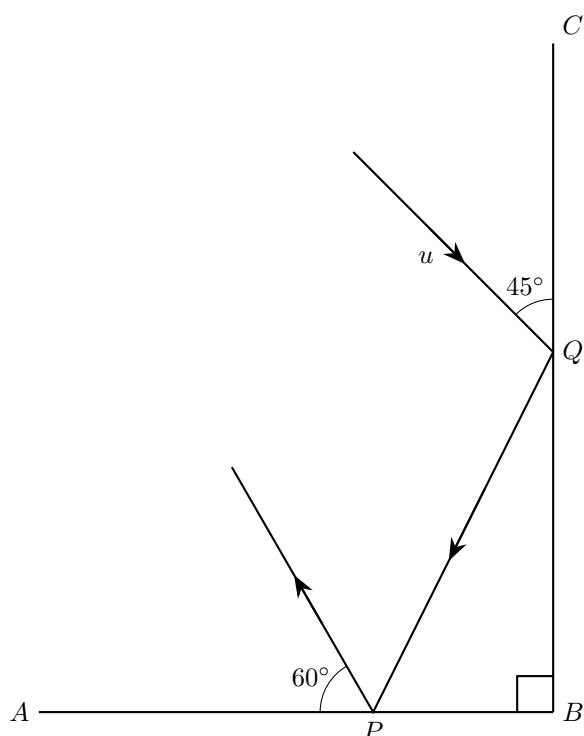
The coefficient of restitution between  $P$  and this wall is  $e$ .

- (a) Find the value of  $e$ . [5]

After this collision,  $P$  goes on to hit a second smooth vertical wall which is parallel to the direction vector  $\mathbf{i} - 2\mathbf{j}$ . The coefficient of restitution between  $P$  and this second wall is  $\frac{1}{2}$ .

The angle through which the direction of motion of  $P$  is deflected by its collision with this second wall is  $\alpha^\circ$ .

- (b) Find the value of  $\alpha$ , giving your answer to the nearest whole number. [5]



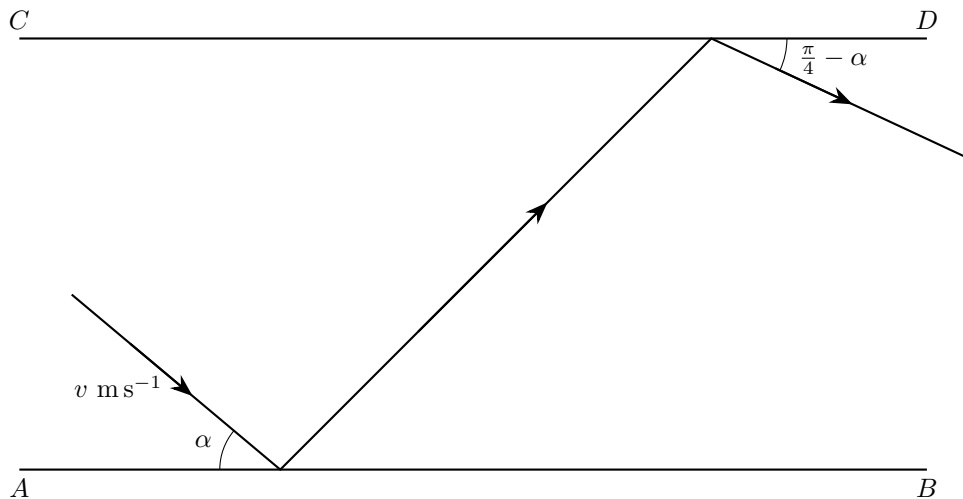
8. The diagram above represents the plan view of part of a horizontal floor, where  $AB$  and  $BC$  are perpendicular vertical walls.  
The floor and the walls are modelled as smooth.

A ball is projected along the floor towards  $BC$  with speed  $u \text{ m s}^{-1}$  on a path at an angle of  $45^\circ$  to  $BC$ . The ball hits  $BC$  at  $Q$  and then hits  $AB$  at  $P$ . After striking  $AB$ , the ball moves away on a path at an angle of  $60^\circ$  to  $AB$ .

The ball is modelled as a particle.

The coefficient of restitution between the ball and wall  $BC$  is  $\frac{1}{2}$ .

- (a) Find the coefficient of restitution between the ball and wall  $AB$  and hence show that, using this model, the final kinetic energy of the ball is 50% of the initial kinetic energy of the ball. [8]
- (b) In practice the walls and the floor may not be smooth. Explain how this model is likely to affect the percentage of kinetic energy remaining. [1]



9. The diagram above represents the plan view of part of a horizontal floor, where  $AB$  and  $CD$  represent fixed vertical walls, with  $AB$  parallel to  $CD$ .

A small ball is projected along the floor towards wall  $AB$ . Immediately before hitting wall  $AB$ , the ball is moving with speed  $v \text{ m s}^{-1}$  at an angle  $\alpha$  to  $AB$ , where  $0 < \alpha < \frac{\pi}{4}$ .

The ball hits wall  $AB$  and then hits wall  $CD$ .

Immediately after the impact with wall  $CD$ , the ball is moving at an angle  $\frac{\pi}{4} - \alpha$  to  $CD$ .

The coefficient of restitution between the ball and wall  $AB$  is  $\frac{3}{5}$ .

The coefficient of restitution between the ball and wall  $CD$  is  $\frac{1}{2}$ .

The floor and the walls are modelled as being smooth. The ball is modelled as a particle.

- (a) Show that

$$\tan \alpha = \frac{2}{3} \quad [7]$$

- (b) Find the percentage of the ball's initial kinetic energy that is lost in the two impacts. [4]

- 10.** A particle  $P$  of mass  $m$  is moving downwards and to the right in a direction making an angle  $45^\circ$  with the horizontal when it strikes a fixed smooth inclined plane. The plane is inclined to the horizontal at an angle  $\alpha$ , where  $0 < \alpha < 45^\circ$ .

At the instant immediately before the impact, the speed of  $P$  is  $u$ .

At the instant immediately after the impact,  $P$  is moving upwards and to the right in a direction making an angle  $45^\circ$  with the horizontal, with speed  $v$ .

- (a) Show that the magnitude of the impulse exerted on the plane by  $P$  is  $mu \sec(45^\circ - \alpha)$ . [5]

The coefficient of restitution between  $P$  and the plane is  $e$ , where  $0 < e < 1$ .

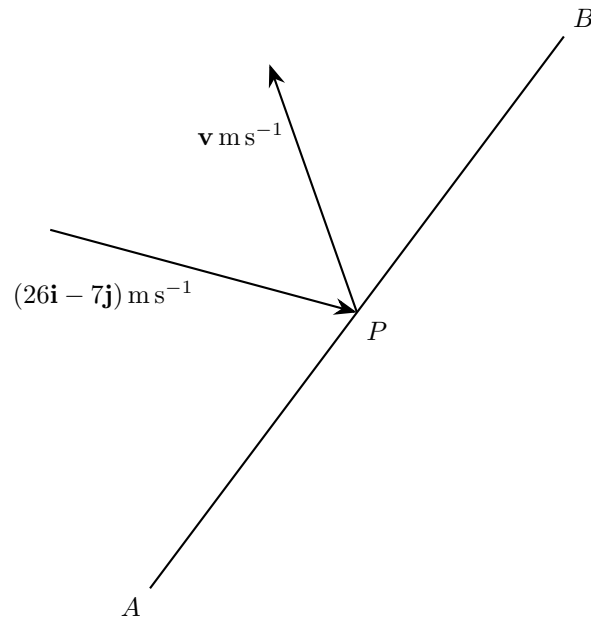
- (b) Show that [3]

$$v^2 = u^2 (\cos^2(45^\circ + \alpha) + e^2 \sin^2(45^\circ + \alpha))$$

- (c) Show that the kinetic energy lost by  $P$  in the impact is

$$\frac{1}{2}mu^2(1 - e^2)\sin^2(45^\circ + \alpha) \quad [2]$$

- (d) Hence find, in terms of  $m$ ,  $u$  and  $e$  only, the kinetic energy lost by  $P$  in the impact. [2]



11. The diagram above represents the plan view of part of a smooth horizontal billiards table, where  $AB$  is a fixed smooth cushion.

The direction of  $\overrightarrow{AB}$  is parallel to the vector  $(3\mathbf{i} + 4\mathbf{j})$ .

A small disc of mass  $0.2 \text{ kg}$  is moving on the table when it strikes the cushion  $AB$ .

Immediately before impact with  $AB$ , the velocity of the disc is  $(26\mathbf{i} - 7\mathbf{j}) \text{ m s}^{-1}$ .

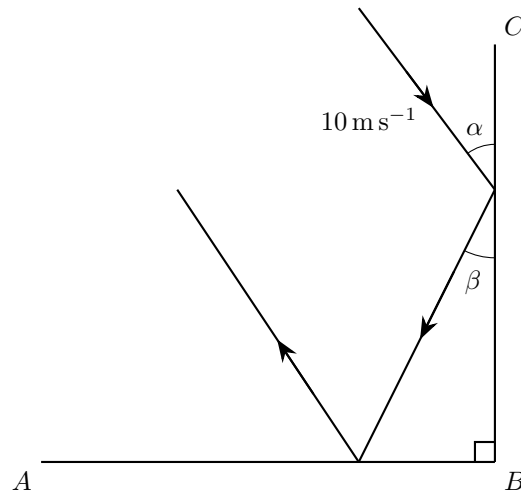
Immediately after impact with  $AB$ , the velocity of the disc is  $\mathbf{v} \text{ m s}^{-1}$ .

The coefficient of restitution between the disc and the cushion is  $\frac{3}{5}$ .

By modelling the disc as a particle,

(a) show that  $\mathbf{v} = -6\mathbf{i} + 17\mathbf{j}$  [6]

(b) Find the magnitude of the impulse exerted by the cushion on the disc in the impact. [3]



- 12.** The diagram above represents the plan view of one corner of a smooth air-hockey table, where  $AB$  and  $BC$  are fixed cushions with  $AB$  perpendicular to  $BC$ .

A small puck is struck along the table towards  $BC$  with speed  $10 \text{ m s}^{-1}$  on a path that makes an angle  $\alpha$  with  $BC$ , where  $\tan \alpha = \frac{3}{4}$ . The puck hits  $BC$  and then hits  $AB$ .

Immediately after hitting  $BC$ , the puck is moving at an angle  $\beta$  to  $BC$ , where  $\tan \beta = \frac{1}{2}$ .

The coefficient of restitution between the puck and  $BC$  is  $e$ .

The coefficient of restitution between the puck and  $AB$  is  $\frac{3}{4}$ .

By modelling the puck as a particle and the table and cushions as being smooth,

- (a) show that  $e = \frac{2}{3}$  [5]
- (b) find the speed of the puck immediately after it hits  $AB$ . [4]
- (c) Suggest two ways in which the model could be refined to make it more realistic. [2]